



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

AIOT

03-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	B	A	B	A	C	C	C	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	D	D	B	B	D	A	D	A	C

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	A	C	C	B	A	D	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	C	A	C	C	C	D	D	A

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	A	D	A	C	D	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	A	B	A	A	B	B	A	A

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (A)**

$$\vec{r}_f = \vec{r}_i + \Delta\vec{s}_1 + \Delta\vec{s}_2 + \dots$$

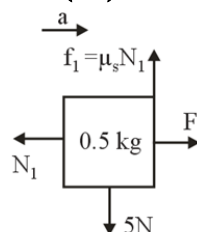
$$\vec{r}_f = (2\hat{i} + 3\hat{j}) + 5\hat{i} + 8\hat{j} + (-2\hat{i} + 4\hat{j}) + (-6\hat{j})$$

$$\vec{r}_f = 5\hat{i} + 9\hat{j}$$

Distance from thrower

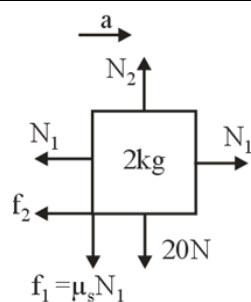
$$= \sqrt{5^2 + 9^2} = \sqrt{106}$$

2. **Ans (C)**



$$F - N_1 = 0.5 a$$

$$f_1 = \mu_s N_1 = 5N$$



$$N_1 - f_2 = 2a$$

$$f_2 = \mu N_2$$

$$N_2 = 20 + f_1 = 25 N$$

$$N_1 - \frac{1}{5}(25) = 2a$$

$$N_1 = 5 + 2a$$

$$\mu_s N_1 = 5$$

$$\mu_s (5 + 2a) = 5$$

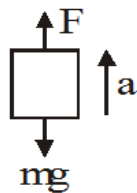
$$\frac{1}{2}(5 + 2a) = 5$$

$$2a = 5$$

$$a = 2.5 \text{ m/s}^2$$

3. **Ans (B)**

by F.B.D of block



$$F - mg = ma$$

$$30(1 - 3y^2) - 20 = 2a$$

$$30 - 90y^2 - 20 = 2a$$

$$10 - 90y^2 = 2a$$

$$a = 5 - 45y^2$$

$$\int_0^0 v dV = \int_0^H (5 - 45y^2) dy$$

$$0 = 5H - 15H^3$$

$$H = \pm \frac{1}{\sqrt{3}}$$

4. **Ans (A)**

Time take by stone to reach floor $t = \frac{h}{u}$

To find the speed of man we can use

momentum conservation

$$Mv_1 = mu$$

$$v_1 = \frac{mu}{M}$$

Distance of man moved above platform is

$$y = \left(\frac{mu}{M}\right) \left(\frac{h}{u}\right) = \frac{mh}{M}$$

$$\text{Total height} = h \left(1 + \frac{m}{M}\right)$$

5. **Ans (B)**

$$U_i = \frac{KQ^2}{2a} + \frac{KQ^2}{4a} + \frac{KQ^2}{2a} = \frac{5KQ^2}{4a}$$

$$\text{and } U_f = \frac{K(2Q)^2}{4a} = \frac{KQ^2}{a}$$

$$\Rightarrow \text{Heat produced} = U_i - U_f = \frac{KQ^2}{4a}$$

6. **Ans (A)**

λ_1 = wavelength of the incident sound

$$= \frac{10u - \frac{u}{2}}{f_1} = \frac{19u}{2f_1}$$

f_1 = frequency of the incident sound

$$= \left(\frac{10u - u}{10u - \frac{u}{2}}\right) f = \frac{18}{19} f$$

$= f_r$ = frequency of the reflected sound

λ_r = wavelength of the reflected sound

$$= \frac{10u + u}{f_r} = \frac{11u}{18f} \times 19 = \frac{11 \times 19}{18} \cdot \frac{u}{f}$$

$$\frac{\lambda_i}{\lambda_r} = \frac{19u}{2f} \times \frac{18f}{11 \times 19u} = \frac{9}{11}$$

8. **Ans (C)**

$$r = R \sin \theta$$

$$\text{Required force} = [2\pi r] T \sin \theta = 2\pi R \sin^2 \theta T$$

9. **Ans (C)**

Since B is grounded, therefore $V_B = 0$

$$\frac{kQ_A}{R} + \frac{kQ_B}{R} = 0 \quad (R = \text{radius of shell B})$$

$$Q_B = -Q_A$$

$$\sigma_B = -Q$$

$$\sigma_B = -\frac{Q}{4\pi R^2}$$

$$\text{But } \sigma_A = \frac{Q}{4\pi r^2}$$

$\therefore$ (A) is not correct

Apply Gauss's theorem. Only (C) is correct

10. **Ans (B)**

$$mg \times 2R = \frac{1}{2}mv^2 + \frac{1}{2} \times mR^2 \times \left(\frac{v}{R}\right)^2$$

$$\Rightarrow v = \sqrt{2gR} = 5 \text{ m/s}$$

11. **Ans (D)**

$$Mg = \int_0^{\ell/4} \rho_0(1 + by)dyAg$$

$$M = \rho_0 A \ell \left[\frac{1}{4} + \frac{b\ell}{16} \right]$$

$$M'g = \int_{\ell/4}^{\ell} \rho_0(1 + by)dyAg$$

$$M' = \rho_0 A \ell \left[\frac{3}{4} + \frac{15}{32}b\ell \right]$$

13. **Ans (D)**

$$\frac{1}{R_{eq}} = \frac{1}{2R} + \frac{1}{R}$$

$$R_{eq} = \frac{2}{3}R$$

$$C_{eq} = \frac{C}{2} + C = \frac{3}{2}C$$

$$\tau = R_{eq}C_{eq} = RC$$

14. **Ans (B)**

$$\frac{q_{in}}{\epsilon_0} = -aR4\pi R^2 + a(2R)4\pi(2R)^2$$

$$q_{in} = 28\pi a\epsilon_0 R^3$$

16. **Ans (D)**

$$\text{Least count} = 0.05 \text{ mm}$$

$$\text{Main scale reading} = 76 \times 0.5 = 38 \text{ mm}$$

$$\text{Vernier scale reading} = 4 \times 0.05 = 0.2 \text{ mm}$$

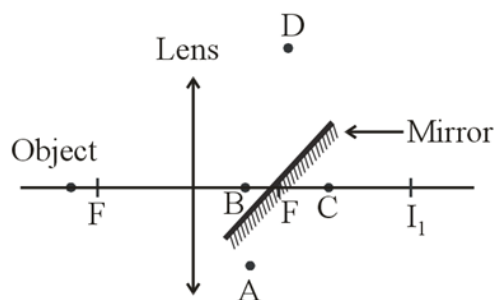
$$\text{Length} = 38.2 \text{ mm} = 3.82 \text{ cm}$$

17. **Ans (A)**

$$\frac{-\left(\frac{dN}{dt}\right)_A}{-\left(\frac{dN}{dt}\right)_B} = 1 = \frac{\lambda_A N_A}{\lambda_B N_B}$$

$$\Rightarrow 1 = \left(\frac{1}{2}\right) \frac{N_A}{N_B} \Rightarrow N_A = 2N_B$$

18. **Ans (D)**



I_1 is the image formed by lens.

I_1 behaves as object for mirror.

Final image is formed at D.

19. **Ans (A)**

$$i_3 = i_1 + i_2 = 5 \left(\frac{3}{5} \sin \omega t + \frac{4}{5} \cos \omega t \right)$$

$$= 5(\sin \omega t \cos 53^\circ + \cos \omega t \sin 53^\circ)$$

$$= 5 \sin (\omega t + 53^\circ)$$

20. **Ans (C)**

Flux can't change in a superconducting loop.

$$\Delta \phi = 2\pi R^2 \cdot B$$

Initially current was zero. So self flux was zero.

$$\therefore \text{Finally } Li = 2\pi R^2 \times B$$

$$i = \frac{2\pi R^2 \times B}{L}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (300)**

$$mg \frac{L}{2} = \frac{1}{2} \left(\frac{mL^2}{3} \right) \omega^2$$

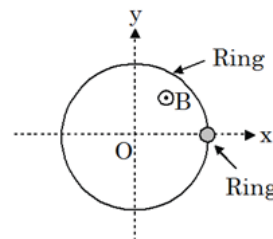
$$\omega = \sqrt{\frac{3g}{L}}$$

Angular impulse about hinge

$$= I\omega = \frac{mL^2}{3} \sqrt{\frac{3g}{L}}$$

2. **Ans (0)**

B depends upon r.



$\therefore$ Flux linked with ring :

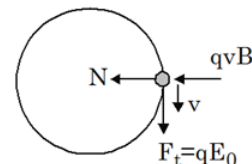
$$\phi = \int_0^R B 2\pi r dr = \int_0^R \frac{E_0 t}{r} 2\pi r dr$$

$$f = 2\pi E_0 t R$$

$$\frac{d\phi}{dt} = 2\pi E_0 R = (\text{induced electric field}) 2\pi R$$

$\therefore$ Induced electric field = E_0 (tangential)

Circular motion of bead:



$$F_1 = qE_0 = \frac{mdV}{dt} \Rightarrow v = \frac{qE_0 t}{m}$$

Centripetal force equation :

$$N + qVB = \frac{mv^2}{R}$$

$$N + q \frac{qE_0 t}{m} \frac{E_0 t}{R} = \frac{m}{R} \frac{q^2 E_0^2 t^2}{m^2}$$

$$N = 0$$

3. Ans (1)

$$v_{B/F} = \frac{d}{dt} \left(y \times \frac{4}{3} + x \right) = \frac{4}{3}(1) + (-1) = \frac{1}{3}$$

$$v_{F/B} = \frac{d}{dt} \left(\frac{x}{4/3} + y \right) = \frac{3}{4}(-1) + 1 = \frac{1}{4}$$

$$\text{Ratio} = \frac{\frac{1}{3}}{\frac{1}{4}} = \frac{4}{3} = \mu n \Rightarrow n = 1$$

4. Ans (3)

For photo-electrons to be emitted,

$$\lambda_{\max} = 10000 \text{ \AA}$$

∴ From Weins displacement law,

$$\lambda_{\max} T \Rightarrow 10,000 \times 10^{-10} \times T = 3 \times 10^{-3}$$

$$\Rightarrow T = 3 \times 10^3 \text{ K}$$

5. Ans (108)

$$P_1^2 = 2m_1 k_1$$

$$P_2^2 = 2m_2 k_2$$

$$\Rightarrow m_1 k_1 = m_2 k_2$$

$$k_2 = \frac{m_1 k_1}{m_2}$$

$$k_2 = \frac{3 \times 216}{6} = 108 \text{ J}$$

6. Ans (1)

$$\text{Given, } \lambda_{\text{wave}} = 3 \times 10^{-5} \text{ m, } \theta = 30^\circ$$

$$d = 0.5 \times 10^{-3} \text{ m}$$

$$D = 1 \text{ m}$$

$$\mu_{\text{glass}} = 1.5$$

$$t_{\text{glass}} = 5 \times 10^{-5} \text{ m}$$

$$\therefore \Delta x \text{ at } 0 = d \sin 30^\circ + t(\mu - 1)$$

$$= 0.5 \times 10^{-3} \times \frac{1}{2} + 5 \times 10^{-5} (0.5)$$

$$= 0.25 \times 10^{-3} + 2.5 \times 10^{-5} = 0.275 \times 10^{-3} \text{ m}$$

$$\therefore \Delta \phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{3 \times 10^{-5}} \times 0.275 \times 10^{-3}$$

$$= \frac{55\pi}{3} = 18\pi + \frac{\pi}{3}$$

∴ For maxima to occur, $\Delta \phi = 0$

⇒ Minimum phase difference to be introduced

$$= -\frac{\pi}{3}$$

$$\therefore y \frac{d}{D} = \frac{\lambda}{3}$$

$$\therefore y = \frac{\lambda D}{6d} = \frac{3 \times 10^{-5} \times 1}{6 \times 0.5 \times 10^{-3}} \text{ m} = 1$$

7. Ans (43)

Angular Momentum about hinge

$$L_i = L_f$$

$$m u \left(\frac{3\ell}{4} \right) = \left(\frac{m\ell^2}{3} + m \left(\frac{3\ell}{4} \right)^2 \right) \omega$$

$$\omega = \frac{36u}{43\ell}$$

8. Ans (1)

$$P = \rho g_{\text{eff}} h$$

$g_{\text{eff}} = 0$ because satellites are under free fall state.

Hence $h = \infty$ but length of tube = 1m

Hence h = can be maximum upto 1m.

9. Ans (3)

$$\Delta Q = \Delta U + W$$

$$W = \frac{1}{4} \Delta Q; W = \frac{\Delta Q}{u}$$

$$\Rightarrow \Delta u = \frac{3}{4} \Delta Q$$

$$nC_v \Delta T = \frac{3}{4} nC \Delta T$$

for process $PV^x = \text{constant}$

$$\Rightarrow C = \frac{4C_v}{3} = C_v + \frac{R}{1-x}$$

$$\frac{R}{1-x} = \frac{C_v}{3} = \frac{R}{2}$$

$$\Rightarrow x = -1$$

$$\Rightarrow PV^{-1} = \text{constant}$$

$$\Rightarrow P^3 V^{-3} = K$$

10. Ans (8)

$$v_1 = \sqrt{\frac{T_1}{\mu_1}} = \sqrt{\frac{2250 \times 10^3}{9}} = 5 \times 100 = 500 \text{ m/s}$$

$$v_2 = \sqrt{\frac{T_2}{\mu_2}} = \sqrt{\frac{2250 \times 10^3}{25}} = 3 \times 100 = 300 \text{ m/s}$$

$$\lambda_1 = \frac{v_1}{f}; \lambda_2 = \frac{v_2}{f}$$

$$\lambda_1 = \frac{1}{3} \text{ m}; \lambda_2 = \frac{1}{5} \text{ m}$$

$$\therefore \text{Net wavelength} = 3 + 5 = 8$$

PART-2 : CHEMISTRY

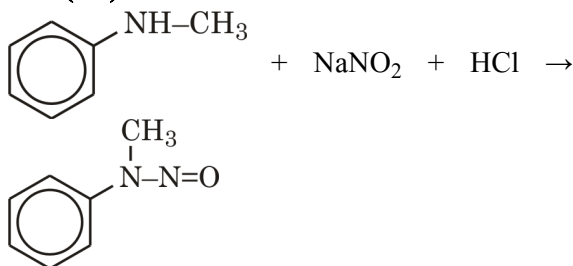
SECTION-I

1. Ans (B)

$$\text{Rate constant } K = \frac{(\text{Rate})}{[\text{conc.}]} = \frac{1 \times 10^{-3}}{.1}$$

$$= \frac{3 \times 10^{-3}}{.3} = 1 \times 10^{-2} \text{ sec}^{-1}$$

2. **Ans (A)**



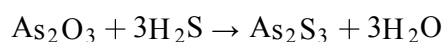
3. **Ans (D)**

Fact based.

4. **Ans (A)**

N-F bond is less polar than N-H bond.

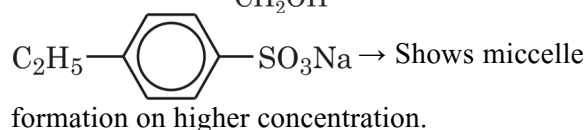
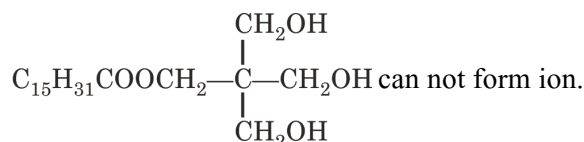
6. **Ans (C)**



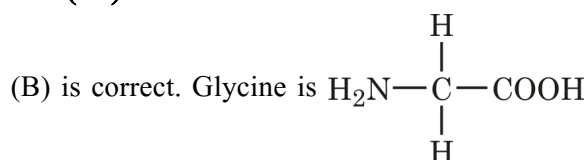
It is double decomposition reaction.

Sulphur sol is likely to be negatively charged due to adsorption of S^{2-} ion.

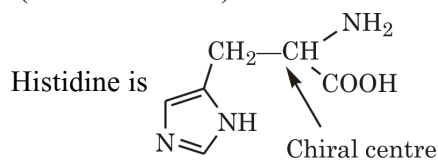
$\therefore \text{Al}^{3+}$ will be most effective.



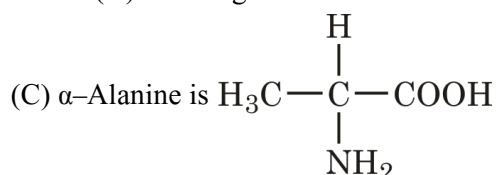
7. **Ans (B)**



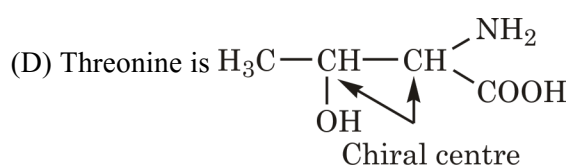
(No chiral centre)



Hence (A) is wrong.



Hence (C) is wrong.



8. **Ans (A)**

$$\pi = \frac{n_B RT}{V}; n_B = \frac{w_B}{M_B}$$

$$\pi = \frac{w_B}{M_B} \times \frac{RT}{V}$$

$$M_B = \frac{w_B}{V} \times \frac{RT}{\pi} = \frac{2 \times 0.0821 \times 300 \times 760}{0.3 \times 20} = 6239.6 \text{ gm mol}^{-1}$$

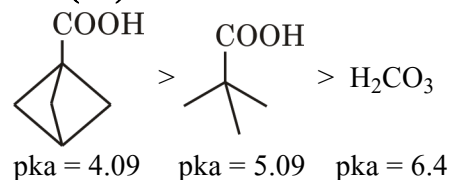
9. **Ans (D)**

End product is isomer of benzene therefore DBE = 4.

10. **Ans (C)**

Aldol condensation.

11. **Ans (A)**

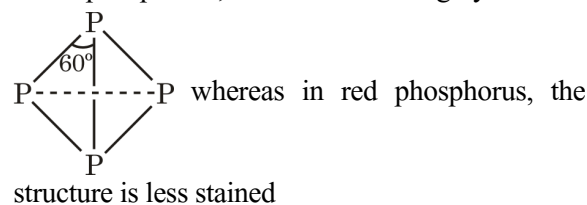


12. **Ans (B)**

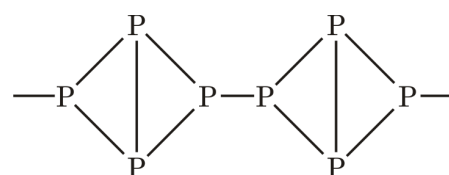
(A) is not correct because it is a correct statement. Nitrogen forms multiple bonds because two N atoms can come sufficiently close together while P atom being large in size cannot come close together and hence does not form $\text{P} \equiv \text{P}$.

(B) is correct because it is an incorrect statement as the P-P single bond is more stronger than N-N because in N-N, there is more repulsive force between the non-bonding electron pair.

(C) is not correct it is a correct statement as in white phosphorus, the structure is highly strained

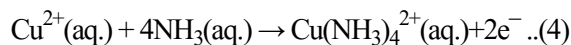
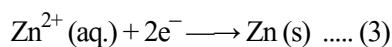
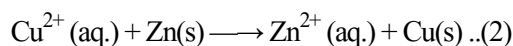
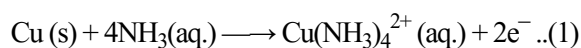


whereas in red phosphorus, the structure is less strained



(D) is not correct because it is a correct statement as S-S single bond is a strong bond.

13. **Ans (C)**



$$(1) = (4) - (2) - (3)$$

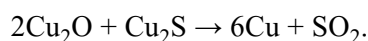
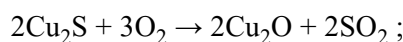
$$-2 \text{FE}^\circ = -RT \ln 10^{20} + 2F(1.1) - 2F(.76)$$

$$E^0 = \frac{2.303RT}{2F} \log 10^{20} - 0.34$$

$$\Rightarrow E^0 = \frac{0.06}{2} \times 20 - 0.34$$

$$= 0.6 - 0.34 = 0.26 \text{ V}$$

14. **Ans (A)**



15. **Ans (C)**

(A) Li $\Rightarrow$ Red, Na $\Rightarrow$ Yellow, K, Rb & Cs

$\Rightarrow$ Blue/Violet

(B) Thermal stability $\propto 1/\phi$

$$D^u \downarrow \rightarrow T. S. \uparrow$$

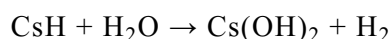
(C) Conc. solution is bronze coloured & diamagnetic.

(D) $D^v \downarrow \rightarrow M - H$ bond becomes weak

$\downarrow$

$$\text{BDE} \downarrow$$

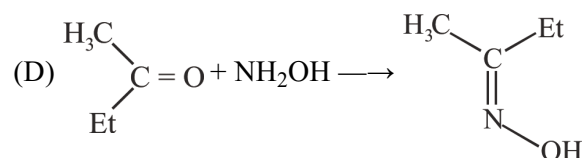
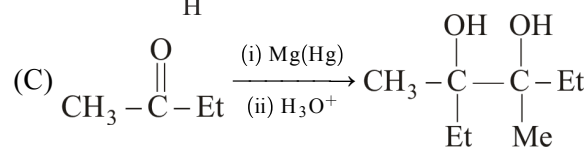
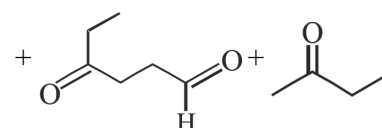
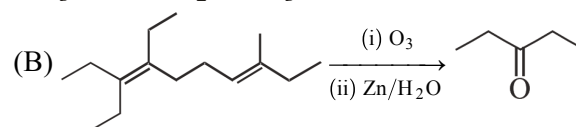
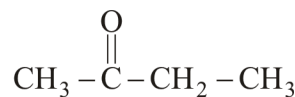
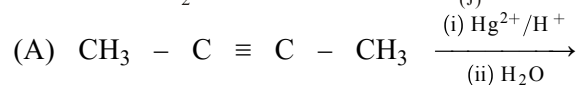
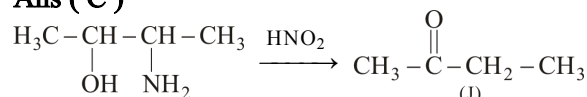
Tendency to reduce other $\uparrow$.



16. **Ans (C)**

In case of (C) option, CH_3O^- is a strong nucleophile in DMSO and confirms $\text{S}_{\text{N}}2$.

17. **Ans (C)**



Can show geometrical isomerism

19. **Ans (D)**

$$E_{\text{Cr}^{3+}/\text{Cr}^{2+}}^0 < E_{\text{Fe}^{3+}/\text{Fe}^{2+}}^0 < E_{\text{Co}^{3+}/\text{Co}^{2+}}^0$$

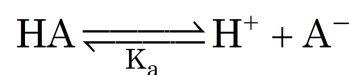
20. **Ans (A)**

Fact based.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5)**



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$\log K_a = \log[\text{H}^+] + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{If } [\text{A}^-] = [\text{HA}] \Rightarrow \text{p}K_a = \text{pH}$$

$$\Rightarrow \text{pH} = 5$$

2. **Ans (2)**

Collision frequency, $Z \propto u_{avg}$ as other factors are constant

$$\text{So, } Z \propto \sqrt{\frac{T}{M}}$$

While, mean free path ' λ '

$\lambda \propto \frac{T}{P}$ but since 'n' & volume are constant, λ is independent of T & M.

Statements (ii) and (vii) are correct.

3. **Ans (6)**

On replacing 'b' by 'a' in $[Mabcdef]^{n\pm}$ the resulting complex will be : $[Ma_2bcde]^{n\pm}$

Total Number of geometrical isomer in $[Mabcdef]^{n\pm} = 15$

Total Number of geometrical isomer in $[Ma_2bcde]^{n\pm} = 9$

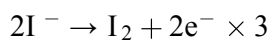
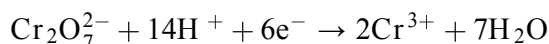
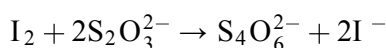
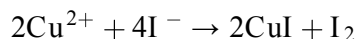
hence number of geometrical isomer reduced = 6

4. **Ans (5)**

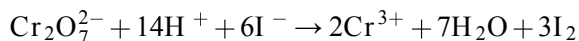
(a), (b), (d), (e), & (h) are incorrect.

5. **Ans (50)**

The reactions involved are:



Combined equation is



$$\therefore \text{Equivalent mass of } \text{K}_2\text{Cr}_2\text{O}_7 = \frac{294}{6} = 49\text{g}$$

$$\text{Normality of } \text{K}_2\text{Cr}_2\text{O}_7 \text{ solution} = \frac{2.45}{49}$$

$$\text{Normality of hypo solution} = \frac{20}{30} \times \frac{2.45}{49}$$

$$\text{Normality of } \text{CuSO}_4 \text{ solution} = \frac{25}{20} \times \frac{20}{30} \times \frac{2.45}{49}$$

$$\text{Mass of Cu in } \text{CuSO}_4 = \frac{25}{20} \times \frac{20}{30} \times \frac{2.45}{49} \times 63 = 2.625\text{g}$$

$$\therefore \% \text{ of Cu in alloy} = \frac{2.625}{5.25} \times 100 = 50\%$$

6. **Ans (3)**

Net work during the cycle

$$W_{\text{net}} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= -P_1 V_1 \ln \frac{P_1}{P_2} - P_2 (V_C - V_B) - P_1 V_2 \ln \frac{P_2}{P_1} - P_1 (V_A - V_D)$$

$$= [P_1 V_2 - P_1 V_1] \ln \frac{P_2}{P_1} - P_2 V_C + P_2 V_B - P_1 V_A + P_1 V_D$$

$$[\text{Also } P_2 V_B = P_1 V_A \text{ and } P_2 V_C = P_1 V_D]$$

$$\therefore |W_{\text{net}}| = P_1 (V_2 - V_1) \ln \frac{P_2}{P_1}$$

$$= 1 \left[\frac{4}{\ln 10} - \frac{1}{\ln 10} \right] \ln \frac{10}{1}$$

$$= \frac{3}{\ln 10} \times \ln 10 = 3$$

7. **Ans (5)**

Energy of photon = 10 eV

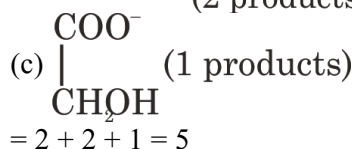
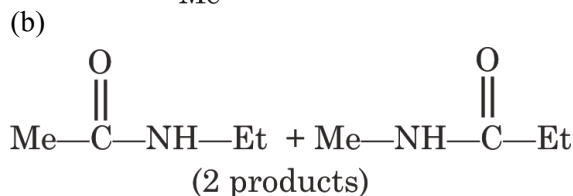
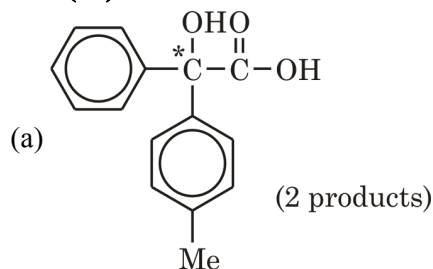
Work function = 4eV

$$\text{K.E.}_{\text{max}} = 6 \text{ eV}$$

$$\lambda = \left(\frac{150}{V} \right)^{1/2} = \left(\frac{150}{6} \right)^{1/2} \text{ \AA}$$

$$= (25)^{1/2} \text{ \AA} = 5 \text{ \AA}$$

8. **Ans (5)**



9. **Ans (2)**

(i), (ii) are correct.

(iii) is Antidepressant.

(iv) is Bacteriocidal.

(v) is Antiseptic.

10. **Ans (6)**

Polystyrene, PVC, Polyacrylonitrile, Teflon, Neoprene & Natural rubber are addition polymer as well as homopolymer.

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

$$f(g(x)) = \sin(\cos x), \text{ period : } 2\pi$$

$$g(f(x)) = \cos(\sin x), \text{ period : } \pi$$

$$f(g(-x)) = \sin(\cos(-x)) = \sin(\cos x) = f(g(x))$$

$$\text{and } g(f(-x)) = \cos(\sin(-x))$$

$$= \cos(-\sin x)$$

$$= \cos(\sin x) = g(f(x))$$

Hence both are even function

2. **Ans (B)**

$$\{x\}^2 + 5m\{x\} - 3m + 1 < 0 \quad \forall x \in \mathbb{R}$$

$$f(t) = t^2 + 5mt - 3m + 1 < 0 \quad \forall t \in [0, 1)$$

$$\Rightarrow f(0) < 0 \Rightarrow m > \frac{1}{3} \quad \dots(A)$$

$$\text{and } f(1) < 0 \Rightarrow 2m + 2 < 0$$

$$\Rightarrow m < -1 \quad \dots(B)$$

from (A) & (B) $m \in \emptyset$

3. **Ans (C)**

$$\text{Let } z = x + iy$$

$$\text{from 2nd equation } x = 6$$

$$3|(x-12) + yi| = 5|x + (y-8)i|$$

$$9[36 + y^2] = 25[36 + (y-8)^2]$$

(substituting $x = 6$)

$$9 \cdot 36 + 9y^2 = 25 \cdot 36 + 25[y^2 + 64 - 16y]$$

$$16y^2 - 25 \cdot 16y + 36 \cdot 16 + 25 \cdot 64 = 0$$

$$y^2 - 25y + 36 + 100 = 0$$

$$y^2 - 25y + 136 = 0$$

$$(y-17)(y-8) = 0$$

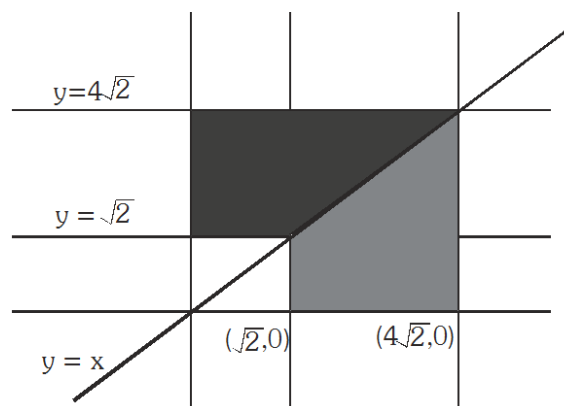
$$\text{then } y = 17$$

$$\text{or } y = 8 \Rightarrow (C)$$

4.

Ans (B)

$$2 \leq \left| \frac{x-y}{\sqrt{2}} \right| + \left| \frac{x+y}{\sqrt{2}} \right| \leq 8 \quad \left. \begin{matrix} x > 0 \\ y > 0 \end{matrix} \right\} \text{ I quad.}$$



$$x \geq y \Rightarrow 2 \leq \frac{x-y}{\sqrt{2}} + \frac{x+y}{\sqrt{2}} \leq 8$$

$$\text{If } x \geq y \text{ then } \sqrt{2} \leq x \leq 4\sqrt{2}$$

$$\text{If } x \leq y \Rightarrow \sqrt{2} \leq y \leq 4\sqrt{2}$$

$$\text{Area} = 32 - 2 = 30$$

5. **Ans (A)**

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} p_1 & q_1 & i \\ p_2 & q_2 & j \\ p_3 & q_3 & k \end{vmatrix}$$

$$[a \ b \ c] \ (p \times q)$$

6.

Ans (D)

Let $P(t^2, 2t)$ is a point on the

parabola then COC is

$$xt^2 + 8ty + 2(x + t^2) + 3 = 0$$

$$\Rightarrow t^2(x+2) + t(8y) + 2x + 3 = 0$$

If chord touches then $D = 0$

$$\Rightarrow \text{curve is } 64y^2 = 4(x+2)(2x+3)$$

$$\Rightarrow 16y^2 - 2x^2 - 7x - 6 = 0$$

$$D \neq 0, h^2 > ab \Rightarrow \text{hyperbola}$$

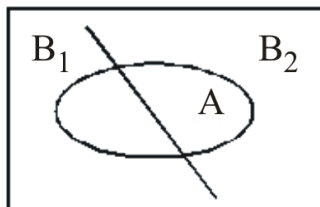
7. **Ans (A)**

A : policyholder has an accident within a year.

B_1 : policyholder is accident prone.

B_2 : policyholder is not accident prone.

$$\text{Given } P(B_1) = \frac{3}{10}; P(B_2) = \frac{7}{10}; P(A/B_1) = 0.6; P(A/B_2) = 0.8$$



$$\begin{aligned} P(A) &= P(A \cap B_1) + P(A \cap B_2) \\ &= P(B_1) P(A/B_1) + P(B_2) P(A/B_2) \\ &= \frac{3}{10} \times \frac{6}{10} + \frac{7}{10} \times \frac{8}{10} = \frac{74}{100} = \frac{37}{50} \\ &= 0.74 \text{ Ans.} \end{aligned}$$

8. **Ans (C)**

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{f(x) \cdot \frac{x^2}{2} + x \int_0^x f(t) dt}{x^2} \\ &= \frac{f(0)}{2} + \lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x} = 1 + f(0) = 3 \end{aligned}$$

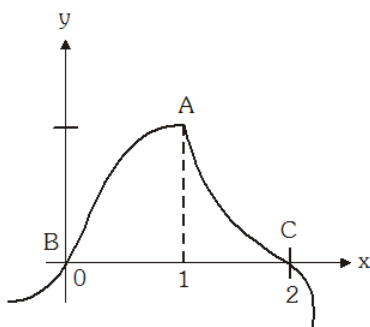
9. **Ans (D)**

By using condition of tangency, we get $4h^2 = 3k^2 + 2$
 $\therefore$ Locus of $P(h, k)$ is $4x^2 - 3y^2 = 2$ (which is hyperbola.)

$$\text{Hence } e^2 = 1 + \frac{4}{3} \Rightarrow e = \sqrt{\frac{7}{3}} \text{ Ans.}$$

10. **Ans (C)**

A, B, C are the 3 critical points of $y = f(x)$



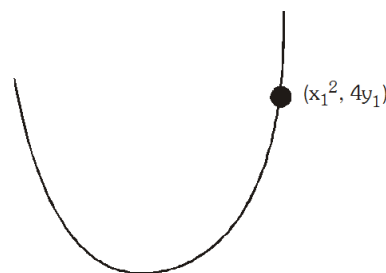
11. **Ans (D)**

$$\sim \{(p \wedge \sim q) \vee q\} = (\sim p \vee q) \wedge (\sim q)$$

12. **Ans (A)**

$$\begin{aligned} & \frac{2^x}{2^2} + \frac{2^x}{2^3} + \frac{2^x}{2^4} + \dots = \sqrt{1 + 3 \cdot 2^{x-2}} \\ & \frac{2^x}{4} \left\{ 1 + \frac{1}{2} + \frac{1}{2^2} + \dots \right\} = \sqrt{1 + 3 \cdot 2^{x-2}} \\ & \frac{2^x}{4} \{2\} = \sqrt{1 + 3 \cdot 2^{x-2}} \\ & \frac{2^x}{2} = \sqrt{1 + 3 \cdot 2^{x-2}} \\ & 2^{2x} = 4 + 12 \cdot \frac{2^x}{2^2} \\ & 2^{2x} - 3 \cdot 2^x - 4 = 0 \\ & p^2 - 3p - 4 = 0 \\ & p(p - 4) + 1(p - 4) = 0 \\ & p = -1, p = 4 \\ & 2^x = -1, \text{ not possible} \\ & 2^x = 4 \\ & x = 2 \end{aligned}$$

13. **Ans (A)**



$$\begin{aligned} 2x &= \frac{4dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{2x_1}{4} \\ \therefore \text{eq. of normal } \frac{y - y_1}{x - x_1} &= \frac{-2}{x_1} \\ \frac{y - \frac{x_1^2}{4}}{x - x_1} &= \frac{-2}{x_1} \text{ pass through } (1, 2) \\ \frac{\frac{8 - x_1^2}{4}}{1 - x_1} &= \frac{-2}{x_1} \Rightarrow \frac{8 - x_1^2}{4(1 - x_1)} = \frac{-2}{x_1} \\ 8x_1 - x_1^3 &= -8 + 8x_1 \\ x_1 &= 2 \quad y_1 = 1 \\ \therefore \frac{y - 1}{x - 2} &= -1 \Rightarrow y - 1 = -x + 2 \\ y + x - 3 &= 0 \end{aligned}$$

14. **Ans (B)**

Let $\frac{x}{c} = z \Rightarrow \frac{1}{c} dx = dz$, when $x \rightarrow ac$, then $z \rightarrow a$ and $z \rightarrow bc$, then $z \rightarrow b$ given integral

$$\int_a^b f(z) dz = \int_a^b f(x) dx$$

PART-3 : MATHEMATICS

SECTION-II

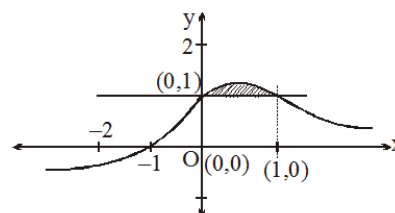
1. **Ans (1)**

On solving C : $y = \frac{x+1}{x^2+1}$ and the line L : $y = 1$,

we have

$$\frac{x+1}{x^2+1} = 1 \Rightarrow x+1 = x^2+1$$

$$\Rightarrow x(x-1) = 0 \Rightarrow x = 0, 1$$



Thus, the points of intersection are at

$x = 0$ and $x = 1$.

$$\begin{aligned} \text{Required area} &= \int_0^1 \left(\frac{x+1}{x^2+1} - 1 \right) dx \\ &= \frac{1}{2} \int_0^1 \frac{2x}{x^2+1} dx + \int_0^1 \frac{1}{x^2+1} dx - \int_0^1 dx \\ &= \left[\frac{1}{2} \ln(x^2+1) + (\tan^{-1} x) - x \right]_0^1 \\ &= \frac{1}{2} \ln 2 + \frac{\pi}{4} - 1 \text{ Ans.} \end{aligned}$$

2. **Ans (196)**

Let $x_i = 2n_i - 1$ where $n_1, n_2, n_3, n_4 \geq 1$

Hence we have

$$(2n_1 - 1) + (2n_2 - 1) + (2n_3 - 1) + (2n_4 - 1) = 98$$

$$n_1 + n_2 + n_3 + n_4 = 51$$

giving n_1, n_2, n_3, n_4 each equal to 1

We have $n_1 + n_2 + n_3 + n_4 = 47$

using beggar $\underbrace{000 \dots 0}_{47} \underbrace{\cancel{0}\cancel{0}\cancel{0}}_3$

$$\Rightarrow {}^{50}C_3 = \frac{50 \cdot 49 \cdot 48}{6} = 19600$$

$$= \frac{n}{100} = 196$$

15. **Ans (A)**

$$\sum_{r=1}^6 r^3 = 441 \text{ if equation has a solution then}$$

$\sin(r\pi x)$ should be equal to 1

for $r = 1, 2, \dots, 6$ which is not possible

16. **Ans (A)**

As we are interested in coefficient of t^{50} , we shall ignore all the term with exponent more than 50.

Thus we can write as

$$(1 + {}^{25}C_1 t^2 + \dots + {}^{25}C_{25} t^{50}) \times (1 + t^{25} + t^{40} + t^{45} + t^{47})$$

As all the terms in the first have even exponent we can ignore t^{25}, t^{45} and t^{47} too thus coefficient of t^{50} is $= {}^{25}C_{25} + {}^{25}C_5 = 1 + {}^{25}C_5$

17. **Ans (B)**

$$2x^3 y dy + 3x^2 y^2 dx = \frac{2y x dx - x^2 dy}{y^2}$$

$$\int d(x^3 y^2) = \int d\left(\frac{x^2}{y}\right)$$

18. **Ans (B)**

radical centre is (3, 2)

19. **Ans (A)**

$$\begin{vmatrix} 2\beta\gamma - \alpha^2 & \gamma^2 & \beta^2 \\ \gamma^2 & 2\alpha\gamma - \beta^2 & \alpha^2 \\ \beta^2 & \alpha^2 & 2\alpha\beta - \gamma^2 \end{vmatrix} = \begin{vmatrix} \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \\ \alpha & \beta & \gamma \end{vmatrix} \begin{vmatrix} \gamma & \alpha & \beta \\ \beta & \gamma & \alpha \\ -\alpha & -\beta & -\gamma \end{vmatrix} = \begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}^2$$

20. **Ans (A)**

$$\sum_{m=1}^n \tan^{-1} \left(\frac{2m}{m^4 + m^2 + 2} \right)$$

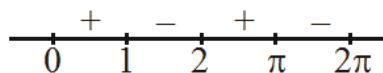
$$= \sum_{m=1}^n \tan^{-1} \left(\frac{(m^2 + m + 1) - (m^2 - m + 1)}{1 + (m^2 + m + 1)(m^2 - m + 1)} \right)$$

$$= \sum_{m=1}^n [\tan^{-1} (m^2 + m + 1) - \tan^{-1} (m^2 - m + 1)]$$

$$= \tan^{-1} (n^2 + n + 1) - \tan^{-1} 1 = \tan^{-1} \left(\frac{n^2 + n}{2 + n^2 + n} \right)$$

3. **Ans (1)**

$$f'(x) = (e^x - 1)\sin x \cdot (x-1)(x+1)(x-2)(x+2)2x$$



$$\therefore L = 2, M = 1$$

4. **Ans (0)**

Angle between tangent from point on director circle is $\frac{\pi}{2}$

5. **Ans (1)**

Required plane

$$(1 + 2\lambda)x + (2 - 3\lambda)y + (3 + 7\lambda) + 5 + \lambda = 0$$

it is parallel to given line $\Rightarrow \sin \theta = 0$

$$8(1 + 2\lambda) - 7(2 - 3\lambda) - 4(3 + 7\lambda) = 0$$

$$\Rightarrow \lambda = 2$$

$$\text{x-intercept} = \frac{-5 - \lambda}{1 + 2\lambda} = -\frac{7}{5}$$

6. **Ans (3)**

by comparison of power

$$x = 3$$

7. **Ans (3)**

$$\text{adj}(A) = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \Rightarrow |\text{adj}A| = 4$$

$$\Rightarrow |A|^2 = 4$$

$$\text{Now, } \therefore A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$\Rightarrow A^{-2} = \frac{(\text{adj}(A))^2}{|A|^2} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}^2$$

$$= \frac{1}{4} \begin{bmatrix} 5 & 6 & 5 \\ 5 & 5 & 6 \\ 6 & 5 & 5 \end{bmatrix}$$

$$\frac{1}{4} \sum_{i=1}^3 \sum_{j=1}^3 x_{ij} = \frac{\text{sum of all the elements}}{4} = 3$$

8. **Ans (6)**

$$f(x) = f(4 - x)$$

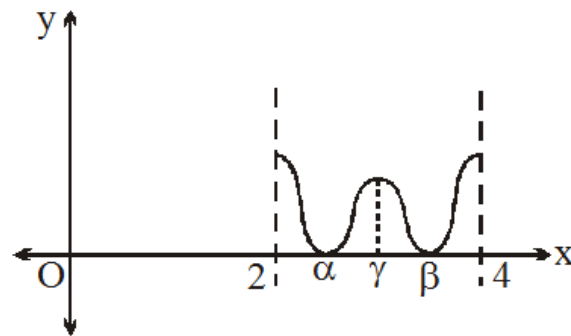
symmetric about $x = 2$

$\therefore$ If $f(x) = 0$ has 2 distinct repeated root in $(2, 4)$

$\Rightarrow f'(x) = 0$ has atleast 3 roots in $(2, 4)$

$\therefore f'(x) = 0$ will have atleast 7 roots in $[0, 4]$

including the local entremum at $x = 2$.



as $f(x)$ is differentiable & continous and symmetric about $x = 2$.

$\therefore f''(x) = 0$ has atleast 6 roots. (by Rolle's) Ans.

9. **Ans (1)**

$$y = x[\ln x - \ln(a + bx)]$$

$$y' = \frac{a}{a + bx} + \frac{y}{x}$$

$$y'' = \frac{-ab}{(a + bx)^2} - \frac{y}{x^2} + \frac{1}{x}y'$$

$$y'' = \frac{a^2}{x(a + bx)^2}$$

$$x^3 y'' = \frac{a^2 x^2}{(a + bx)^2}$$

$$x^3 y'' = (xy' - y)^2$$

10. **Ans (22)**

$$S(x_i - \bar{x})^2 = 648$$

$$\sigma^2 = \frac{648}{8} = 81$$

$$\sigma^2 = 81$$

$$\sigma = 9$$

$$\text{coefficient of variation is} = \frac{9}{40} \times 100 = \frac{45}{2} = 22.5\%$$